

Random Forest Algorithms :-

Introduction:-

Random Forest is a widely used machine learning algorithm developed by Leo Breiman and Adele Cutler that combines the results of multiple decision trees to obtain a single result. Its ease of use and flexibility have fueled its adoption, as it handles both classification and regression problems.

What is Random forest ?

Random forest- like group decision making in machine learning. We know that a forest has countless trees, and where there are many trees, the forest is very big. Similarly, the more trees there are in the random forest algorithm, the higher its accuracy and problem-solving ability. Random forest is a classification system that consists of multiple decision trees on various subsets of the data set and averages them to increase prediction accuracy for that data. It is based on the concept of ensemble learning, which is the process of combining multiple classifiers to solve complex problems and improve model performance.

What is Random Forest Algorithm?

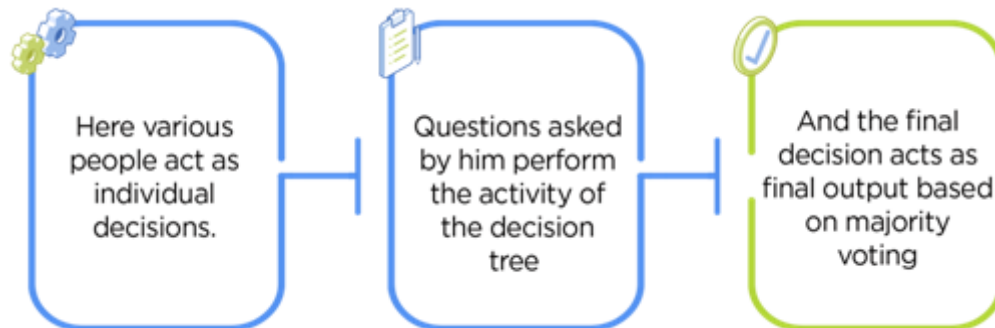
The popularity of the random forest algorithm is due to its user-friendliness and flexibility, which allows it to solve classification and recovery problems. The power of this algorithm lies in its ability to handle complex data and reduce overfitting; this makes it useful for many prediction tasks in machine learning.

One of the most important features of the random forest algorithm is that it can process data containing continuous variables (such as regression) and categorical variables (such as distribution). It is more efficient in distribution and recycling. In this article, I will try to make you understand how Random Forest works and how to use Random Forest in classification.

Real-Life Analogy of Random Forest

Let's dive into the real-life analogy of Random Forest to better understand its concept. Student named X wants to choose courses after 10+2 and is confused about choosing courses according to his ability. So, he decided to consult many people, including his sibling, teachers, parents, college

students, and employees. Ask them why they should choose the department, career reasons, tuition fees, etc. He asked many questions. Finally, after consulting other people about education, he decided to take classes that most people recommended.



Working of Random Forest Algorithm

Before understanding how random forest algorithm works in machine learning, we need to learn the machine learning tool. Integration simply means combining multiple models. So, a set of models is not a single model that can make predictions.

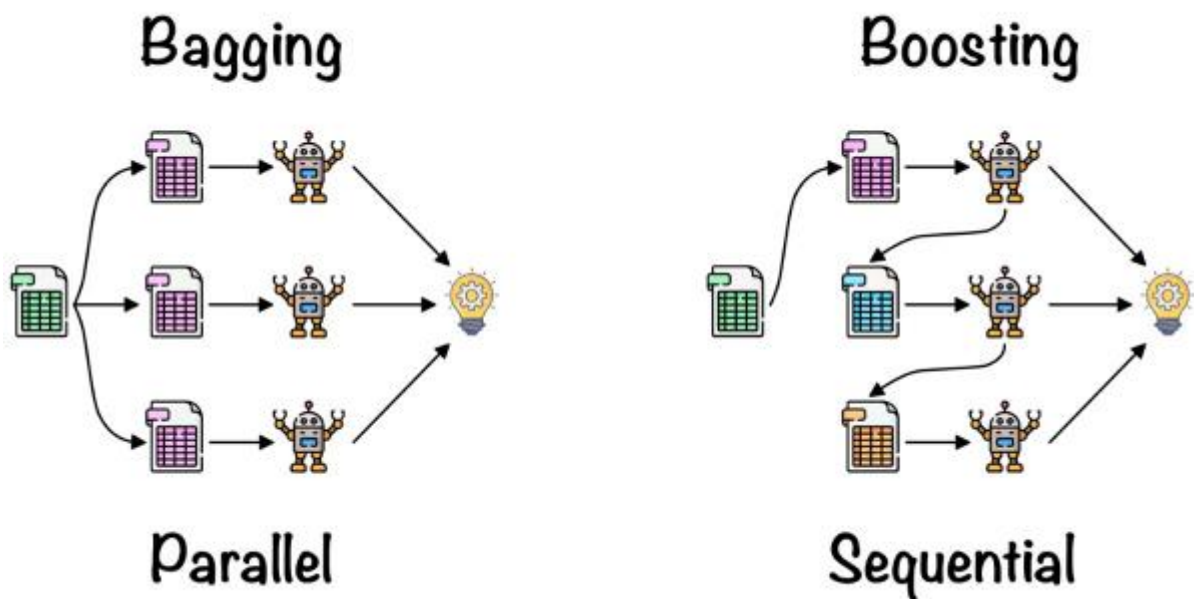
Ensemble uses two types of methods:

Bagging:-

Creates different training methods by changing the training data model and the final output is based on majority voting. For example, Random Forest.

Boosting:-

Turn weak students into strong students by creating a combination of models so that the final model has the highest accuracy. For example, ADA BOOST, XG BOOST.

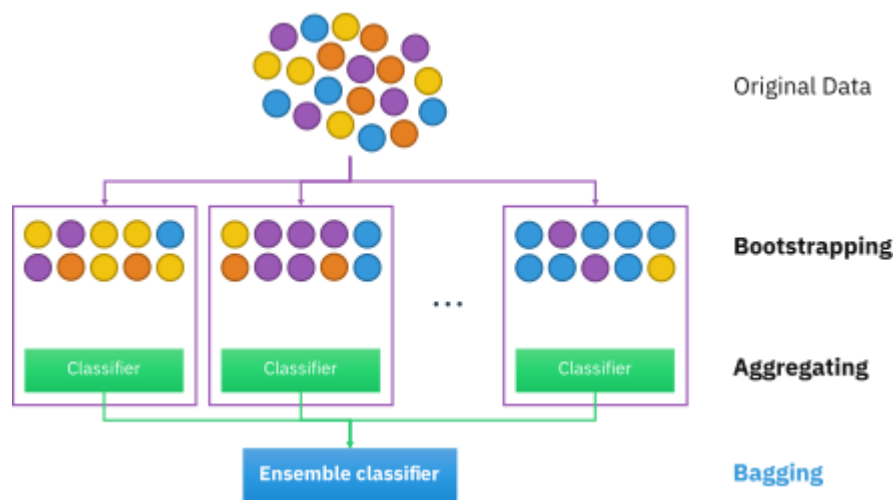


As mentioned before, Random Forest works according to the Bagging principle. Now let's move on to the details of bagging.

Bagging

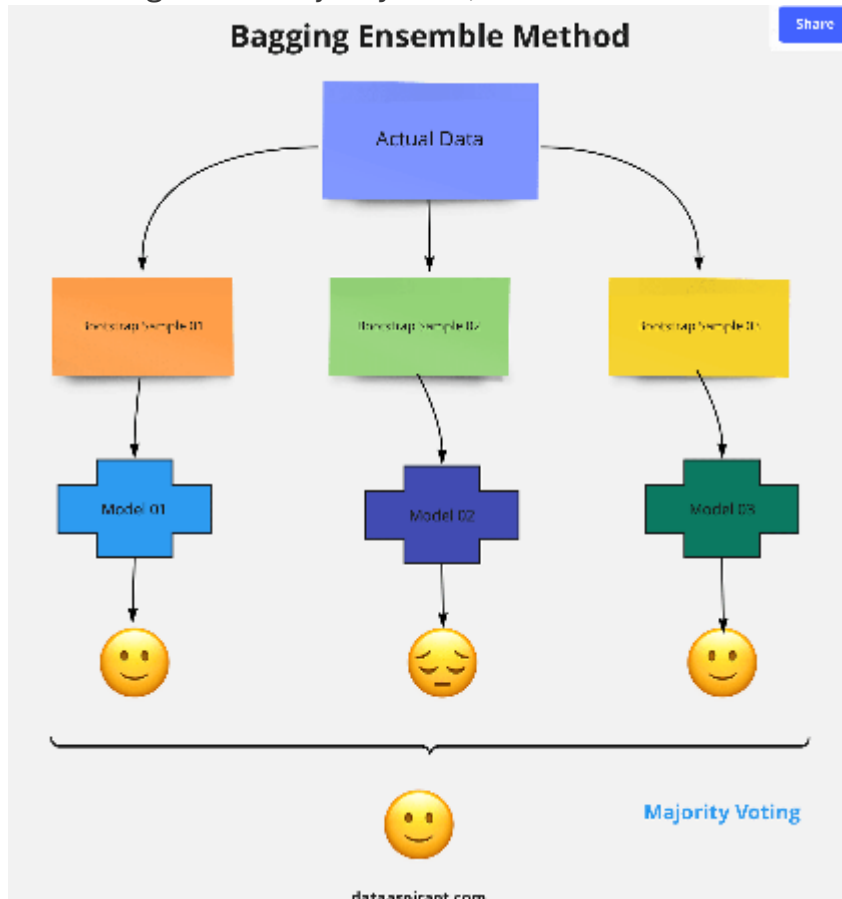
Bagging, also known as Bootstrap Aggregation, serves as the ensemble technique in the Random Forest algorithm. Here are the steps involved in Bagging:

1. **Selection of Subset:** Bagging begins by selecting a sample or subset from the entire dataset.
2. **Bootstrap Sampling:** Each sample is then based on this sample, called the bootstrap sample, which is taken from the original data and modified. This process is called linear sampling.
3. **Bootstrapping:** The linear sampling or row sampling step with replacement is called bootstrapping.
4. **Independent Model Training:** Each model is trained independently from bootstrap samples. This training process produces results for each model.
5. **Majority Voting:** The final result is determined by combining all models by majority vote. Choose the best guess from the model.
6. **Aggregation:** This step involves combining all the results and creating the final result based on the majority vote; This is called aggregation.



Now let's break down the example to see it with the help of the diagram below.

These Bootstrap samples are taken from real data (Bootstrap Sample 01, Bootstrap Sample 02 and Bootstrap Sample 03) with modifications; This means that each instance will generally not contain any private data. The models obtained from these bootstrap models (Model 01, Model 02, and Model 03) were examined independently. All design results are shown in the figure. Nowadays, happy emojis are often compared to sad emojis. Therefore, according to the majority vote, the latest version is the happy emoji.



Boosting

Boosting is a method in which the concept of learning is used together. Boosting algorithms combine various simple models (also called weak learners or basic estimators) to produce the final result. It is done by creating a model using a weak model in series.

There are many support algorithms; AdaBoost is the first best boost and is designed for binary distribution. AdaBoost is short for Adaptive Boosting. It is a popular boost that turns many "weak classifiers" into "strong classifiers". There are other developments as well.

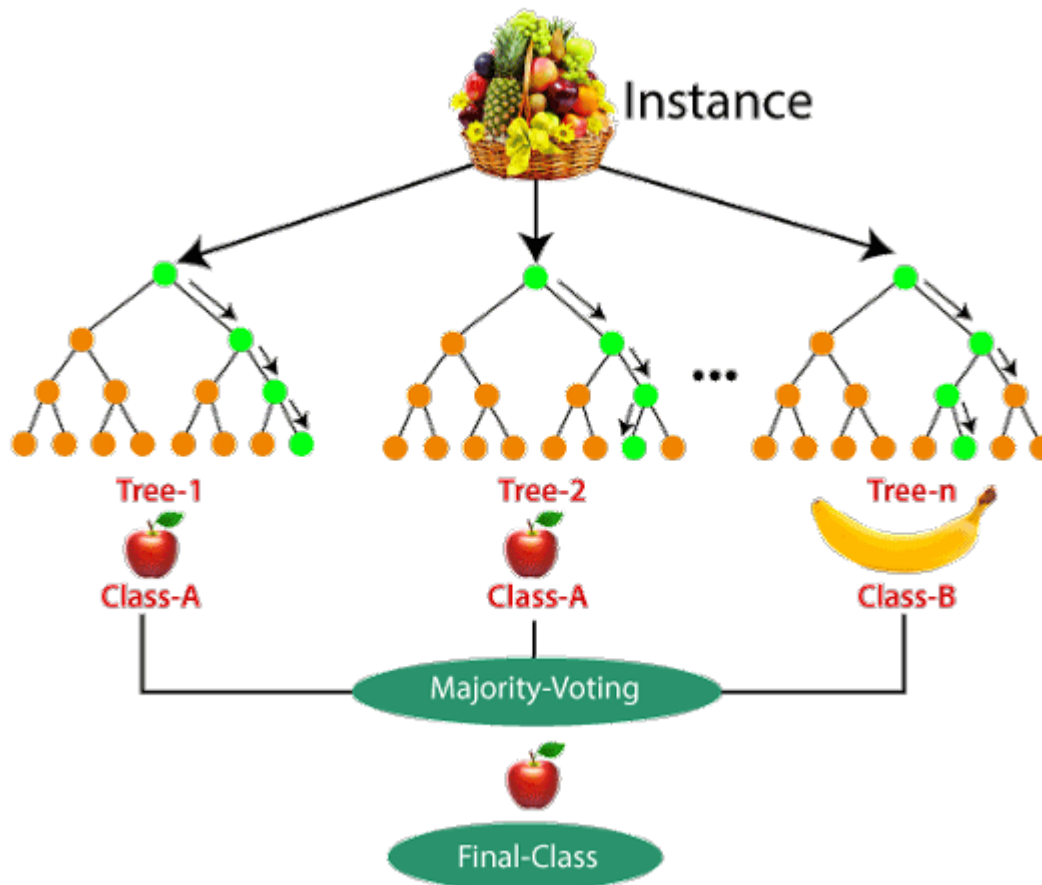
Steps Involved in Random Forest Algorithm

- **Step 1:** In the Random Forest model, a group of data points and features are selected by the process to build each decision tree. Simply put, from a dataset containing k data, n random records and m features are obtained.
- **Step 2:** Individual decision trees are constructed for each sample.
- **Step 3:** Each decision tree will generate an output.
- **Step 4:** Final output is considered based on **Majority Voting or Averaging** for Classification and regression, respectively.

For example

Consider the fruit basket as the data as shown in the figure below. Now n number of samples are taken from the fruit basket, and an individual decision tree is constructed for each sample. Each decision tree will generate an output, as shown in the figure. The final output is considered based on majority voting.

In the below figure, you can see that the majority decision tree gives output as an apple when compared to a banana, so the final output is taken as an apple.



Important Features of Random Forest

- **Diversity:** Not all attributes/variables/features are considered while making an individual tree; each tree is different.
- **Immune to the curse of dimensionality:** Since each tree does not consider all the features, the feature space is reduced.
- **Parallelization:** Each tree is created independently out of different data and attributes. This means we can fully use the CPU to build random forests.
- **Train-Test split:** In a random forest, we don't have to segregate the data for train and test as there will always be 30% of the data which is not seen by the decision tree.
- **Stability:** Stability arises because the result is based on majority voting/averaging.

Difference Between Decision Tree and Random Forest

Random forest is a collection of decision trees; still, there are a lot of differences in their behavior.

Decision trees	Random Forest
1. Decision trees normally suffer from the problem of overfitting if it's allowed to grow without any control.	1. Random forests are created from subsets of data, and the final output is based on average or majority ranking; hence the problem of overfitting is taken care of.
2. A single decision tree is faster in computation.	2. It is comparatively slower.
3. When a data set with features is taken as input by a decision tree, it will formulate some rules to make predictions.	3. Random forest randomly selects observations, builds a decision tree, and takes the average result. It doesn't use any set of formulas.

Thus, random forests are much more successful than decision trees only if the trees are diverse and acceptable.

Important Hyperparameters in Random Forest

Hyperparameters are used in random forests to either enhance the performance and predictive power of models or to make the model faster.

Hyperparameters to Increase the Predictive Power

- **n_estimators:** Number of trees the algorithm builds before averaging the predictions.
- **max_features:** Maximum number of features random forest considers splitting a node.
- **mini_sample_leaf:** Determines the minimum number of leaves required to split an internal node.
- **criterion:** How to split the node in each tree? (Entropy/Gini impurity/Log Loss)
- **max_leaf_nodes:** Maximum leaf nodes in each tree

Hyperparameters to Increase the Speed

- ***n_jobs***: it tells the engine how many processors it is allowed to use. If the value is 1, it can use only one processor, but if the value is -1, there is no limit.
- ***random_state***: controls randomness of the sample. The model will always produce the same results if it has a definite value of random state and has been given the same hyperparameters and training data.
- ***oob_score***: *OOB* means out of the bag. It is a random forest cross-validation method. In this, one-third of the sample is not used to train the data; instead used to evaluate its performance. These samples are called out-of-bag samples.

Random Forest Algorithm Use Cases

This algorithm is widely used in E-commerce, banking, medicine, the stock market, etc.

For example: In the Banking industry, it can be used to find which customer will default on a loan.

Advantages and Disadvantages of Random Forest Algorithm

Advantages

- It can be used in classification and regression problems.
- It solves the problem of overfitting as output is based on majority voting or averaging.
- It performs well even if the data contains null/missing values.
- Each decision tree created is independent of the other; thus, it shows the property of parallelization.
- It is highly stable as the average answers given by a large number of trees are taken.

- It maintains diversity as all the attributes are not considered while making each decision tree though it is not true in all cases.
- It is immune to the curse of dimensionality. Since each tree does not consider all the attributes, feature space is reduced.
- We don't have to segregate data into train and test as there will always be 30% of the data, which is not seen by the decision tree made out of bootstrap.

Disadvantages

- Random forest is highly complex compared to decision trees, where decisions can be made by following the path of the tree.
- Training time is more than other models due to its complexity. Whenever it has to make a prediction, each decision tree has to generate output for the given input data.